UNIT
1

THE CELL CYCLE, MITOSIS, AND MEIOSIS

Learning objectives:

- Describe the role of chromosomes in inheritance and cell division.
- Understand the cell cycle and explain the stages of Interphase (G1, S, G2).
- Identify and explain the stages of mitosis in creating identical daughter cells.

Introduction to the building block of Life

All living things are made up of tiny units called cells. Cells are the building blocks of life, just like bricks are the building blocks of a house. There are many different types of cells, but all of them have some common parts, such as:

1. Cell Membrane : This is the outer layer that protects the cell and controls what goes in and out.

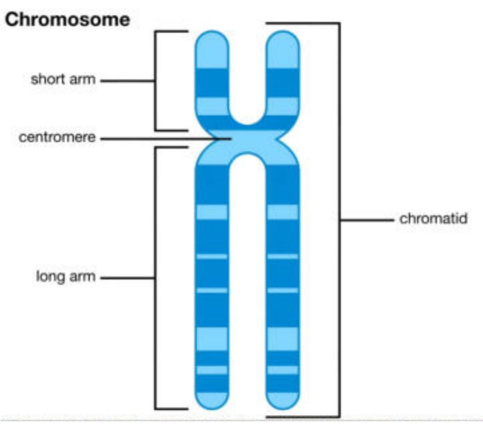
2. Cytoplasm : A jelly-like substance inside the cell where all the other parts float.

3. Nucleus : This is the control center of the cell. It tells the cell what to do and how to grow. Inside nucleus there are chromosomes which play a vital role in developing of new cell.

1.1 Chromosomes

Structure of Chromosomes

Chromosomes are thread-like structures made up of DNA (Deoxyribonucleic Acid)and protein. They are found inside the nucleus of a cell. Each chromosome consists of two sister chromatids, which are joined at a region called the centromere. The structure of chromosomes allows them to carry genetic information in the form of genes, which determine an organism's traits.



Chromosomes

Functions of Chromosomes

Chromosomes carry genetic information like eye colour, height and blood group type that is essential for growth, development, and functioning. During cell division, they ensure that each new cell receives a complete set of genes. Chromosomes also play a crucial role in inheritance, passing traits from parents to offspring.

1.2 Cell Cycle

Introduction to the Cell Cycle

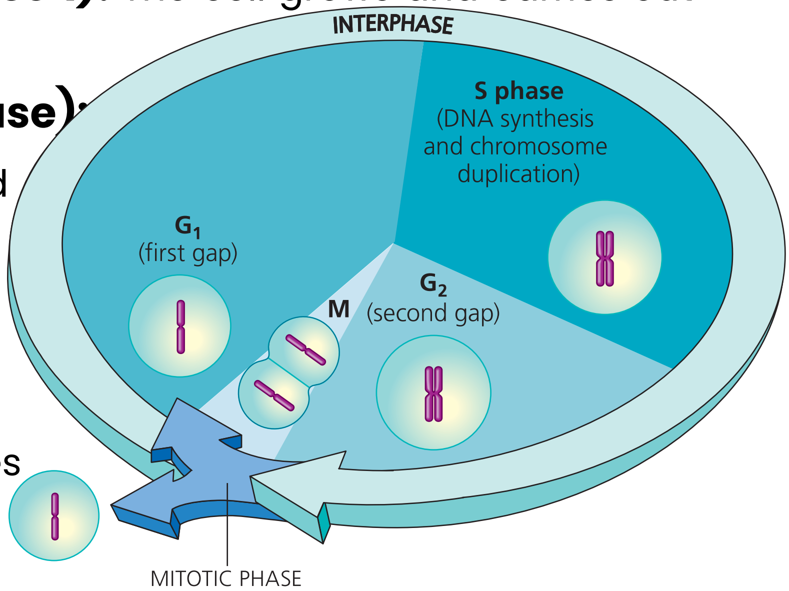
The cell cycle is the series of events that occur in a cell as it grows and divides. It consists of two main stages: Interphase and the Mitotic phase.

During the interphase, the cell prepares for division, and during the mitotic phase, the cell divides into two identical daughter cells.

Interphase

Interphase is the longest phase of the cell cycle and can be broken down into three stages:

- **G1 Phase (Growth Phase 1):** The cell grows and carries out normal functions.
- **S Phase (Synthesis Phase):** DNA replication occurs, and the cell's chromosomes duplicate.
- **G2 Phase (Growth Phase 2):** The cell continues to grow and prepares for

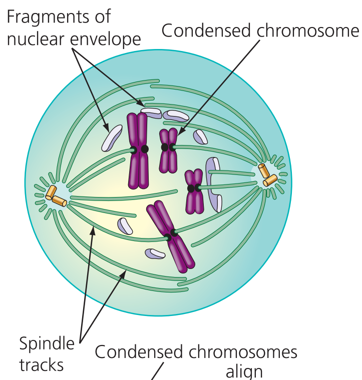


1.3 Mitosis

Overview of Mitosis

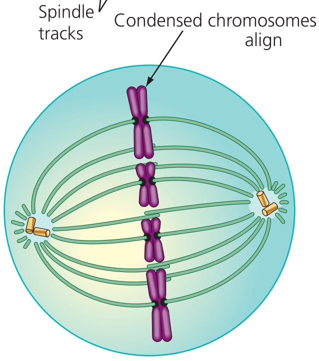
Mitosis is the process by which a single cell divides into two identical daughter cells. It ensures that each daughter cell receives an exact receives an exact copy of the original cell's DNA. Mitosis is important for growth, tissue repair, and asexual reproduction.

Phases of Mitosis



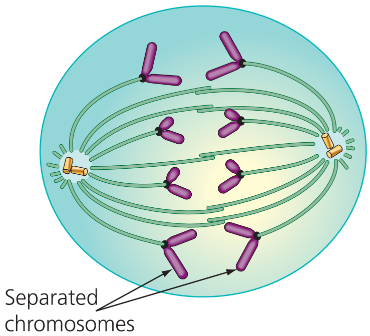
Mitosis occurs in several stages:

1. Prophase: The chromatin condenses into visible chromosomes, and the nuclear membrane begins to break down.



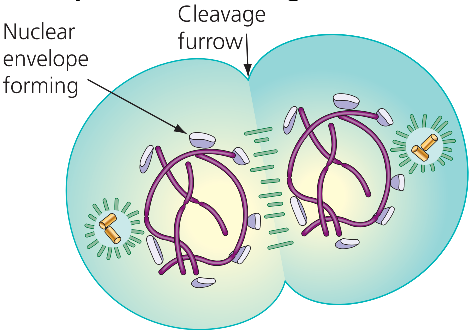
2. Metaphase: The chromosomes align in the middle of the cell.

3. Anaphase: The sister chromatids are pulled apart toward opposite ends of the cell.



4. Telophase: The chromatids reach the poles, and new nuclear membranes form around the sets of chromosomes.

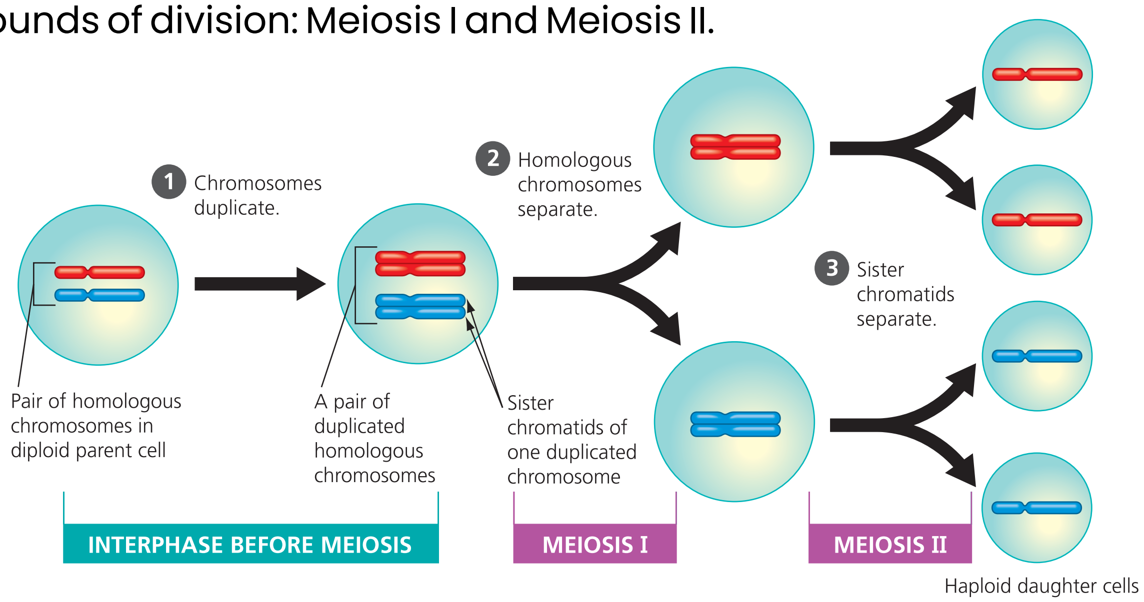
5. Cytokinesis: The cytoplasm divides, resulting in two separate daughter cells.



1.4 Meiosis

Introduction to Meiosis

Meiosis is a type of cell division that reduces the chromosome number by half, resulting in four non-identical daughter cells. It is responsible for the formation of gametes (sperm and egg cells) in animals, and pollen and ovules in plants. Meiosis involves two rounds of division: Meiosis I and Meiosis II.



Meiosis I:

1. Prophase I: Chromosomes get ready, and pairs of chromosomes exchange some DNA (this creates variety).

2. Metaphase I: Chromosome pairs line up in the middle.

3. Anaphase I: The chromosome pairs are pulled apart to opposite sides.

4. Telophase I: Two new cells form, each with half the chromosomes.

5. Cytokinesis: The cell splits into two, each with half the chromosomes.

Meiosis II:

Prophase II: Chromosomes get ready again.

1. **Metaphase II:** Chromosomes line up in the middle (like mitosis).

2. **Anaphase II:** Chromatids are pulled apart to opposite sides.

3. **Telophase II:** Four new cells are formed, each with half the chromosomes.

4. **Cytokinesis:** The cell splits again, making four cells in total.

In Summary:

- **Meiosis I:** It cuts the chromosome number in half.
- **Meiosis II:** Splits the chromosomes into individual parts to make 4 new cells

Importance of Mitosis and Meiosis

Cells reproduce and make more of themselves through two main processes: Mitosis and Meiosis. Each has a unique role in living organisms.

Mitosis

Mitosis is the process by which a single cell divides to produce two identical daughter cells. This is important because:

1. **Growth and Development:** Mitosis helps organisms grow by increasing the number of cells. For example, when you grow taller, it's because your cells are dividing through mitosis.

2. **Repair and Healing:** When you get a cut, mitosis produces new cells to replace the damaged ones and heal the wound.

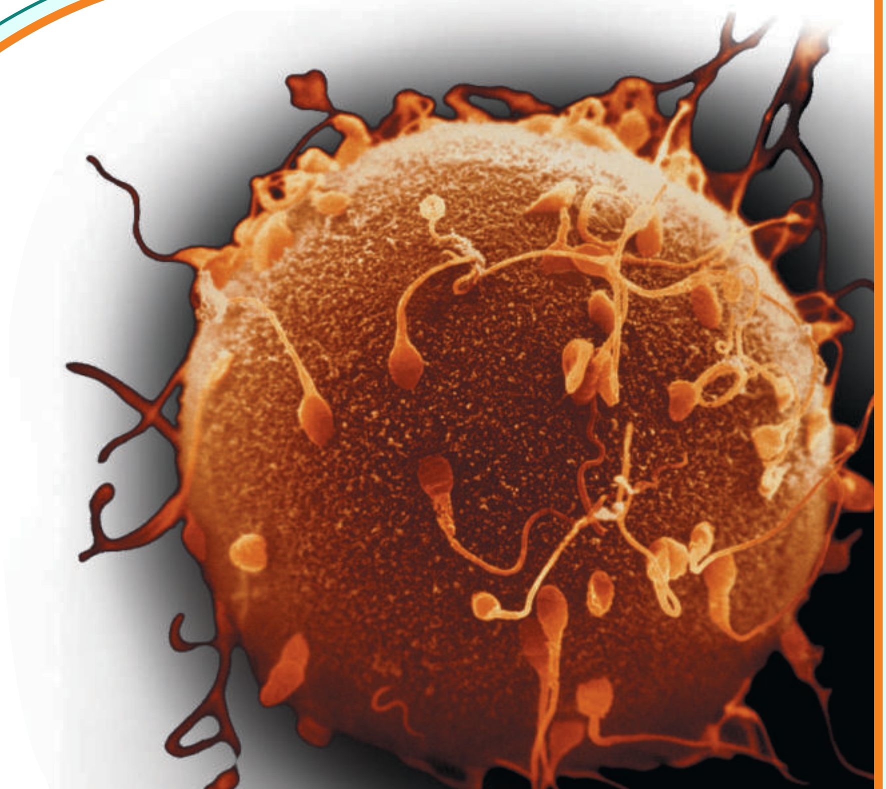
3. **Maintenance:** Mitosis replaces old or worn-out cells, like skin cells that shed every day.

Meiosis

Meiosis is the process that produces sex cells (sperm and eggs) with half the number of chromosomes. This is important because:

1. Genetic Variation: Meiosis creates unique combinations of genes, ensuring each offspring is different from their parents and siblings. This diversity is important for the survival of species.

2. Reproduction: Meiosis produces sex cells (gametes) with half the number of chromosomes (23 in humans). When a sperm and egg combine during fertilization, they form a new cell with the full set of chromosomes (46 in humans).



EXERCISES

Fill in the blanks.

1. Chromosomes are made up of _____________.

2. The main function of chromosomes is to carry _____________.

3. The cell cycle consists of the _______________ phase and the _____________ phase.

4. DNA replication occurs during the _____________ phase.

5. In mitosis, the chromosomes align in the _____________ phase.

6. Meiosis reduces the chromosome number by _____________.

7. _____________ is the phase in which the cell grows and performs its normal functions.

8. The sister chromatids are separated during the _____________ phase of mitosis.

Tick(✓) for True and cross (x) for False statements-

1. The interphase includes the G1, S, and G2 phases.
2. Meiosis results in two identical daughter cells.
3. During prophase, the nuclear membrane forms around the chromosomes.
4. Chromosomes are made up of RNA.
5. The S phase is where the cell prepares for division.
6. Cytokinesis occurs before the telophase.
7. Mitosis is involved in the formation of gametes.
8. Meiosis involves two rounds of division.

Choose the correct answer.

1. What is the structure of chromosomes made up of?

a) Salt b) DNA

c) Water d) Lipids

2. Which phase of the cell cycle is the longest?

a) G1 Phase b) S Phase

c) G2 Phase d) Interphase

3. During which phase of mitosis do the chromosomes align at the centre of the cell?

a) Prophase b) Metaphase

c) Anaphase d) Telophase

4. What is the main purpose of meiosis?

a) Growth and repair b) Formation of gametes

c) DNA replication d) Cell division

5. In which phase do the chromatids separate?

a) Prophase b) Metaphase

c) Anaphase d) Telophase

6. Which of the following is not a phase of mitosis?

a) Interphase b) Prophase

c) Metaphase d) Anaphase

7. What happens during the G2 phase of the cell cycle?

a) DNA replication b) Growth and normal functions

c) Cell prepares for division d) Chromosomes align

8. How many daughter cells are produced in meiosis?

a) Two b) Four

c) Six d) Eight

Answer the Following Questions:

Short-Answer Questions.

1. What is a cell?

2. Write the main parts of the cell?

3. What is the difference between mitosis and meiosis?

Long-Answer Questions.

1. Describe the structure of chromosomes and explain their role in inheritance.

2. What is cell cycle? Explain its stages.

Short Note

1. Write a short note on importance of mitosis and meiosis.

Reasoning question Answers-

1. Why do cells need to grow before they divide?

____

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2. Why do old or damaged cells need to be replaced?

____

____

3. Why can't all cells divide at the same time?

____

____

4. Why is meiosis important for reproduction?

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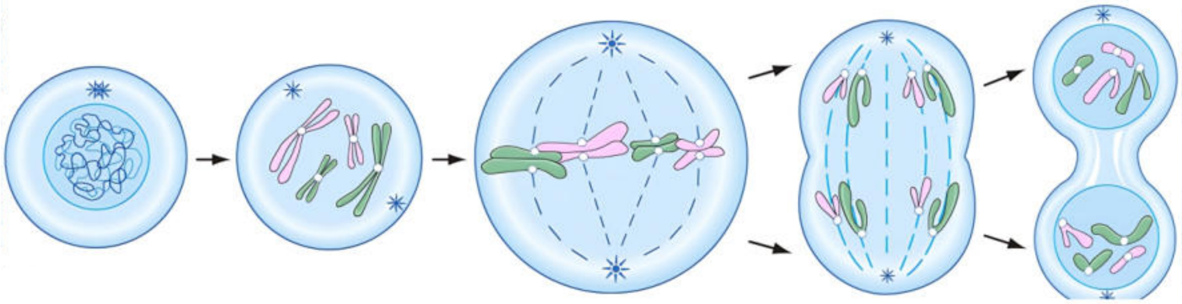
Critical Thinking

1. What would happen if a cell skipped the G1 or G2 phase before mitosis? How would it affect the cell division process?

2. If a cell did not copy its DNA correctly, what could happen to the new cells?

Labelling the Stages of Mitosis

Direction: Label each phase of mitosis and describe what takes place at each phase.



______ ______ ______ ______ ______

Describe what takes place in each phase.

Prophase	
Metaphase	
Anaphase	
Telophase	

MITOSIS AND MEIOSIS

1. Describe 2 parts of meiosis that are similar to mitosis!

a. ______________________________

b. ______________________________

2. Identify whether each process below occurs during mitosis, meiosis, or both!

a. Sister chromatids separate: _________________________

b. Cell division occurs once: _________________________

c. Homologous chromosomes pair: _____________________

d. Crossing over occurs: _________________________

e. Cell division occurs twice: _________________________

f. Replicated chromosomes line up in the middle of the cell: _________________________

3. Does the parent cell in mitosis start off as diploid or haploid?

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